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02455 EXPERIMENT IN COGNITIVE SCIENCE

Final Report: Manipulation of Belief

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1 Introduction

Advances in generative AI have led to a surge in AI-produced visual contents, including images and videos [1]. AI tools can offer creative production [2], [3] but also raise concerns about AI-generated visual misinformation and declining trust in media [4], [5], [6]. As synthetic and real contents become visually indistinguishable, people rely more on cognitive expectations, particularly their beliefs about the nature of the contents [7], [8].

Since people's attitudes towards AI-generated content vary widely [8], their responses are shaped not only by the stimulus but also by top-down processes such as prior beliefs and preferences. Several studies have shown that these beliefs influence emotional reactions. For example, AI-generated charity advertisements elicit lower empathy [9], AI-produced artworks are often judged as less authentic [10], and content labelled as AI-generated receives more negative engagement on social media [11]. These findings highlight how perceptions change when the origin/nature of content is emphasized.

Some studies manipulated authenticity beliefs while keeping the stimuli constant. When participants viewed real emotional faces but were told some were AI-generated, they produced weaker emotional responses and reduced early neural activity [8]. Similarly, labeling emotional images as "real," "AI-generated," or "uncertain" changed physiological signals even when self-reported ratings remained unchanged [12]. Together, these results indicate that beliefs about content authenticity can affect people's response independently of the actual nature of the visual stimulus.

However, the above existing work, which is relevant to belief manipulation about visual content authenticity, has focused almost entirely on static images. Although emotionally positive videos are central to modern communication and misinformation [13], [14], they still remain underexplored. Because videos contain more contexts and change over time, it is unknown whether labeling video as real or AI-generated will influence their reactions in the same way as with images.

To address this gap, we manipulate authenticity beliefs while keeping the video stimuli constant. All participants watch the same real videos but receive different authenticity labels, allowing us to isolate the effect of believing a video is AI- or human-generated on emotional and physiological responses.

Understanding these effects has both practical and theoretical significance. If beliefs strongly shape reactions to video content, then inaccurate labeling or misinformation could distort engagement, trust, and emotional impact. These insights can help guide the strategic use of AI-generated media in organizational or commercial settings. The study also contributes to psychological theory by examining how top-down beliefs affect perception and emotional processing.

Our main research question for this experiment is:

- What are the impacts of prior beliefs about the authenticity of videos (AI-generative vs. human-generative) on people's emotional response?

The hypothesis in this project is:

- Labeling human-made content as 'AI-generated' has an effect on self-reported emotion (valence, arousal and dominance) and heart rate compared to labeling it as 'human-generated'.

2 Methods

2.1 Experimental design

The experiment follows a within-subjects design in which all participants watch the same four videos (A, B, C, D) and each video is paired with an authorship label ("AI-generated" or "Human-made"). Label assignments are counterbalanced so that half of the participants are told that A and B are human-made and C and D are AI-generated, whereas the other half receive the opposite assignments. It ensures that each video appears equally often under both label conditions, see Figure 1.

We selected four animal-themed videos (dog, ants, otter, squirrel) that might elicit positive affect while avoiding ethically inappropriate negative videos. They are provided in A.1 with full credits to

original uploaders. Choosing the same theme of animals helps ensure that observed differences are due to the authorship manipulation rather than people's preferences on contents.

To avoid systematic order effects, the four videos are presented in random sequences. Because the videos were intentionally selected to be highly similar in content, video order is not treated as a separate counterbalancing dimension.

The independent variable is the authorship label (AI vs. Human). The dependent variables are subjective emotional ratings collected via the Self-Assessment Manikin (SAM, including valence (pleasant-unpleasant), arousal (calm-excited), dominance (in control-controlled)) and physiological arousal measured through heart rate monitoring.

2.2 Ethical considerations

The main goal of our experiment is to measure how prior belief about a video's nature affects emotional arousal. To avoid influencing participants' responses, we do not fully disclose our actual study aim. Instead, we simply tell them that we are measuring their reactions to watching videos. After the experiment, all participants will be debriefed and informed about the true purpose and our lies.

2.3 Participants

In order to figure out the sufficient sample size, we looked at previous literature. Since there was no literature about the emotional arousal of AI regarding videos, we could not get an expected effect value for a power analysis. We then looked at articles exploring the relationship between emotional arousal and AI with other medias like images. Articles [8] [12] showed that they employed sample sizes around 30 participants. Thus, we concluded that 30 participants would be sufficient for our experiment as well.

Since we wanted to randomize the video order and stratify the assumed label, we realized that we would only have a limited number of combinations that we could use in order to get unique experiment combinations (as explained in chapter 2.1). By calculating all possible permutations of four videos under two label conditions, we found that exactly 24

unique combinations were available, see Table 1.

Recruitment was carried out among fellow university students, mainly by asking people nearby to participate and asking our acquaintances. Inclusion criteria was adults of age 18 to 60 and fluent in the language of the stimuli. Exclusion criteria was having any conditions that could affect the results of the experiment such as ADHD or heart disease. We also excluded people with a pacemaker or heart transplant, since the provided Firstbeat ECG-sensor documentation stated that this was not allowed.

We ended up using only 22 participants. Initially we had done 24 experiments, but the data for participant 9 (missing data) and 24 (not using the correct experiment design in table 1 by the experimenter's mistakes) were not usable for analysis, hence why we decided to leave it out of the statistics.

2.4 Experimental procedure

All the subjects have been initially informed (see A.5) about the scope of the project, without mention of the manipulation of belief. After reading and signing the documents in A.9 and A.10, the HR sensor was applied. The experiment took place in a PsychoPy interface we created, where four videos of fixed length were shown, with the Self-Assessment Manikin and focus questions in between them. Finally, they were inquired about their general attitude towards AI generated content and their trust on the shown labels. A debriefing about the manipulation of belief completed the trial. The step-by-step procedure can be found in A.4.

2.5 Pre-processing

Since the FirstBeat sensor may continue recording even after it is removed, we rely on each participant's documented start and end timestamps (see step 2 in A.4) to extract the relevant heart-rate segment. For each participant, there is one individual csv file containing this person's heart rate data.

Then, we extracted the data from the PsychoPy files for each participant and put it in one .json file in order to structure the data. For each participant, we paired up the timestamps of each video they watched with the part of their heart rate data that matched.

Based on this we had the RR-intervals for each video separately, with labels. This can be seen in Figure 3. After inspecting the heart rate visualizations, we found that participant 9 had partially missing data and was therefore excluded.

2.6 Data analyses

The data analyses were done on both the heart rate data and SAM data separately, in order to perform statistical tests on the data.

2.6.1 Heart Rate Variability (HRV)

In order to do data analyses and other statistical analyses, we wanted to measure the heart rate variability (HRV) by calculating the root mean square of successive differences (RMSSD) values for each video per participant. The heart rate data with RR-intervals was in time domain, hence why we could calculate RMSSD directly without resampling. We get the RMSSD between normal heartbeats by computing the successive difference between each heartbeat in milliseconds. All of those differences are then averaged and squared in order to get the final RMSSD value [15]. The formula for this calculation can be seen in Equation 1.

$$RMSSD = \sqrt{\frac{\sum_{i=1}^{N-1} (NN_i - NN_{i+1})^2}{N - 1}} \quad (1)$$

NN_i is the i^{th} normal-to-normal heartbeat interval, and N is the total number of NN intervals.

The normal range of HRV for adults is typically around 20 to over 70 ms at rest [16]. Furthermore a lower RMSSD is typically an indicator of stress or fatigue, while a higher RMSSD is typically an indicator of relaxation. RMSSD is inversely related to emotional arousal, where high arousal typically corresponds to a lower RMSSD and vice versa.

In order to find out what statistical test to use on the HRV data, we wanted to first check if it was normal distributed, since it is the most general assumption about parametric tests. We did this by averaging the two "AI-generated" videos with the two "Human-generated" videos per participant, and

then calculated the difference between these two. We then had one difference per participant, and 22 differences in total. We then did a Shapiro-Wilk test on the RMSSD data to check for normality. The Shapiro-Wilk test statistic W is calculated as seen in Equation 2.

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (2)$$

The data is then sorted where $x_{(1)}$ is the smallest value and $x_{(n)}$ is the largest [17]. How close the W statistic is to 1, shows how close the data is to be following a perfect normal distribution. Additionally to the Shapiro-Wilk test we also plotted the data in a histogram and fitted it to a normal distribution, in order to check for normality in a more visual way.

2.6.2 Self-Assessment Manikin (SAM)

We extracted the three SAM responses (valence, arousal, dominance) for each video. These were in the form of discrete ordinal values from 1 to 5. Similarly to 2.6.1, the values of each SAM response have then been averaged by belief across the participants.

Thus, no further statistical analysis is performed on this variable because it is discrete and ordinal, proving additional assumptions unnecessary.

2.7 Statistics

2.7.1 Wilcoxon Signed-Rank Test

Since the HRV differences were not normally distributed (as shown by the Shapiro-Wilk test in Chapter 3.2) and the SAM responses are ordinal, we used the Wilcoxon signed-rank test to compare the AI- and Human-generated conditions for HRV and the three SAM dimensions. This non-parametric test is suitable for within-subjects designs as it compares paired observations from the same participant.

For each dependent variable (RMSSD, valence, arousal, dominance), the paired differences between belief conditions were ranked by absolute value. The Wilcoxon W statistic reflects the sum of the signed ranks and indicates the direction and magnitude of the differences. From this, we obtained the p-values using a 5% significance level. We also computed the

effect size r , shown in Equation 3:

$$r = \frac{z}{\sqrt{n}} \quad (3)$$

where n is the number of paired observations and z is the standardized test statistic. The effect size provides a measure of how consistently participants rated one belief condition higher or lower than the other [18].

2.7.2 Bootstrapping and Confidence Intervals

To expand the results of 2.7.1, we aimed to acquire a confidence interval of 95% by bootstrapping. In practice, data from RMSSD, SAM, and their respective differences by belief, have been empirically extracted from a pool of 10.000 samples randomly selected (with replacement) by the bootstrapping method. The median of the true and bootstrapped samples have then been compared, while also plotting a 95% confidence interval of the bootstrapped pool for the mentioned statistical comparison.

3 Results

3.1 Data distributions

Figure 4 shows a raincloud plot comparing RMSSD between the two belief conditions. It can be seen that the medians are approximately the same, shown by the black line in the middle of the boxes. The whiskers of the two conditions overlap almost completely, with similar tendencies and spread.

Figure 5 shows a boxplot comparing the three SAM responses between the two belief conditions. The whiskers seem to overlap in all cases, with dominance covering a wider range of scores while the others tend to be skewed towards higher or lower values. At the same time, the box sizes and the medians seem to differ between the conditions in each category: Human-Generated content appears to cause significantly higher valence, lower arousal and around the same level of dominance as its counterpart.

3.2 RMSSD: Shapiro-Wilk test

The Shapiro-Wilk Test statistic on the RMSSD data was calculated to be 0.8167, meaning that the data was not significantly close to being normal distributed, but there is still a noticeable deviation. The Shapiro-Wilk test also gave a p-value of 0.0009 at a significance niveau of 0.05, which means that normality can be rejected. Furthermore a distribution of the data can be seen in Figure 2 in the Appendix. This shows that when fitting a normal distribution to the data, it does not follow normality, since the data does not completely match with the blue bell curve generated by the mean and standard deviation of the data.

3.3 Wilcoxon Signed-rank test on RMSSD and SAM

The results of the Wilcoxon Signed-rank test can be seen in Table 2. We used a significance niveau of 5% for all tests. The p-value for the HRV data is $p = 0.4434$, meaning that there is not a statistically significant difference in the HRV data when watching presumed "AI-generated" videos and "human-generated" videos. This is supported by the 95% confidence intervals of "AI" ([15.473, 60.439]) and "Human" ([18.141, 52.567]) overlapping. Like with HRV, arousal and dominance also did not show any statistically significant difference.

The only test statistic that shows a significant difference in values is the valence part of the SAM questionnaire. This has a p-value of $p = 0.0123 < 0.05$ meaning that there is a statistically significant difference at a significance level of 0.05. The effect size of $r = -0.64$ shows that the participants systematically rated a lower valence when watching "AI-generated" videos compared to watching "Human-generated" videos. The negative effect size is because the differences were calculated as '*AI*' - '*Human*'. The 95% CI's does not overlap, which further proves that there is a significant difference between the two. Valence also has the lowest W test statistic, meaning that the "Human-generated" videos were rated higher in valence than "AI-generated" ones.

4 Discussion

It is interesting that only valence showed a significant difference, while the other measures did not. This suggests that participants' subjective positive feelings were influenced by whether they believed the videos were AI- or human-generated, even though their physiological responses were unchanged. The animal videos likely elicited only mild arousal, which may explain the absence of HRV effects. Still, the lower valence ratings for AI-labeled videos indicate that the belief manipulation impacted participants' emotional evaluation.

One of the limitations of the experiment design is that there was no official dataset of videos for arousal experiments we could use, so we had to manually search for videos online and judge ourselves whether they were difficult to distinguish as AI- or human-generated. The lack of a validated video dataset that is known for eliciting arousal and manipulating beliefs makes this experiment less convincing. This limitation suggests why the effects were more subtle than expected.

When we looked at the notes that we took in the debriefing with each participant, we found that around 20 out of 22 people did not believe the labels of the videos when we talked to them in the debriefing conversation. However only 2 participants realized that all videos were real, meaning that most thought that the labels were either switched or that the belief about the videos were being manipulated in some other way. A table of the notes that we took during our conversations can be seen in Table 3 (note that not everything the participants said was written down). This skepticism further showed us that the manipulation may not have worked as intended.

Something else we also realized while doing the experiment was that the video with the ants were very different from the others, in that it was very fast paced with ants running up and down an anthill. This could result in higher arousal when watching this video compared to the other more slow-paced videos, regardless of the label.

These limitations and confounds in our experiment could be why our hypothesis can not be ac-

cepted completely, based on the Wilcoxon signed-rank tests being done in chapter 3.3. Before doing the experiment we made a research question where we expected that there would be an effect in emotional valence and arousal when watching AI-generated videos compared to human-generated videos. Since we only found that there was a statistically significant difference in the SAM response for valence, we can partly accept the hypothesis that there was an effect of participants generally feeling more happy (high end of the valence scale) when watching human-generated videos. However we can not accept the part about an effect in arousal or dominance, since both the SAM arousal/dominance and HRV showed us that there is no statistical significance in arousal or dominance when watching a video with one label compared to the other.

The visualizations in the report support these findings and limitations: the raincloud plot in Figure 4 shows highly overlapping RMSSD distributions across belief conditions, indicating no physiological effect of the manipulation, whereas the SAM distributions suggest clearer subjective differences. Figure 7 shows that participants' prior attitudes toward AI were generally neutral, which may have reduced the impact of the labels. This could be due to our pool of participants consisting of mostly students at DTU, who are most likely familiar with AI. Finally, Figure 6 indicates similar focus levels across conditions, although slightly higher focus for AI-labeled videos may reflect participants attempting to spot AI-related errors, which was evident from the conversations we had with the participants (Table 3).

5 Conclusion

Our results show that the belief manipulation only influenced valence, with participants rating videos as more positive when they believed they were human-generated. No effects were found for arousal, HRV or dominance. Limitations such as the small sample size, failed manipulated belief about the labels, and variability in the stimuli likely contributed to this outcome. Therefore, the hypothesis is only partially accepted.

A Appendix

A.1 Sources

All the sources for the experiment setup, the PsychoPy code to run it, the raw resulting data and the Python code to analyze it, are publicly available in this [GitHub repository](#).

A.2 About LLM usage

In the perspective of fair and ethical usage of LLMs, we state that the usage ChatGPT and Copilot has been limited to support tool for writing and debugging the source code and to correct grammatical mistakes in the text. In writing the present document, we felt the commitment to maintain the genuineness of our handwritten text.

As an example, the following prompts have been sent to LLMs as part of our workflow:

- Prompt: "Can you check whether the texts have grammar mistakes? (...)"
- Prompt: "The spaces between headings and texts are kinda large, how can I make them smaller?"
- Prompt: "Do you know The root mean square of successive differences (RMSSD)? Can you explain to me how to use it when dealing with heart rate? What kinds of values can we get for using RMSSD with RR interval? And if it is a good indicator to run the Shapiro test"
- Prompt: "how to make this table more narrow `\begin{tabular*}{\textwidth}{...}`?"
- Prompt: "change it to a subplot-style where the x axis is in common and contains only the values 1-5"
- Prompt: "remove this file and add the equivalent sample visualization as first cell of the visualization.ipynb file where you just plot the participant1.csv file data in an equivalent form as done in this file. make the code the simplest to understand"
- Prompt: "can you help me write some code to extract the psychopy data from all the participant sv files"
- Prompt: "Can you help me make a data structure of our HRV data in Python? i want it to be in a json file"
- Prompt: "can you make a table for these (...) four wilcoxon tests"

A.3 Experiment design

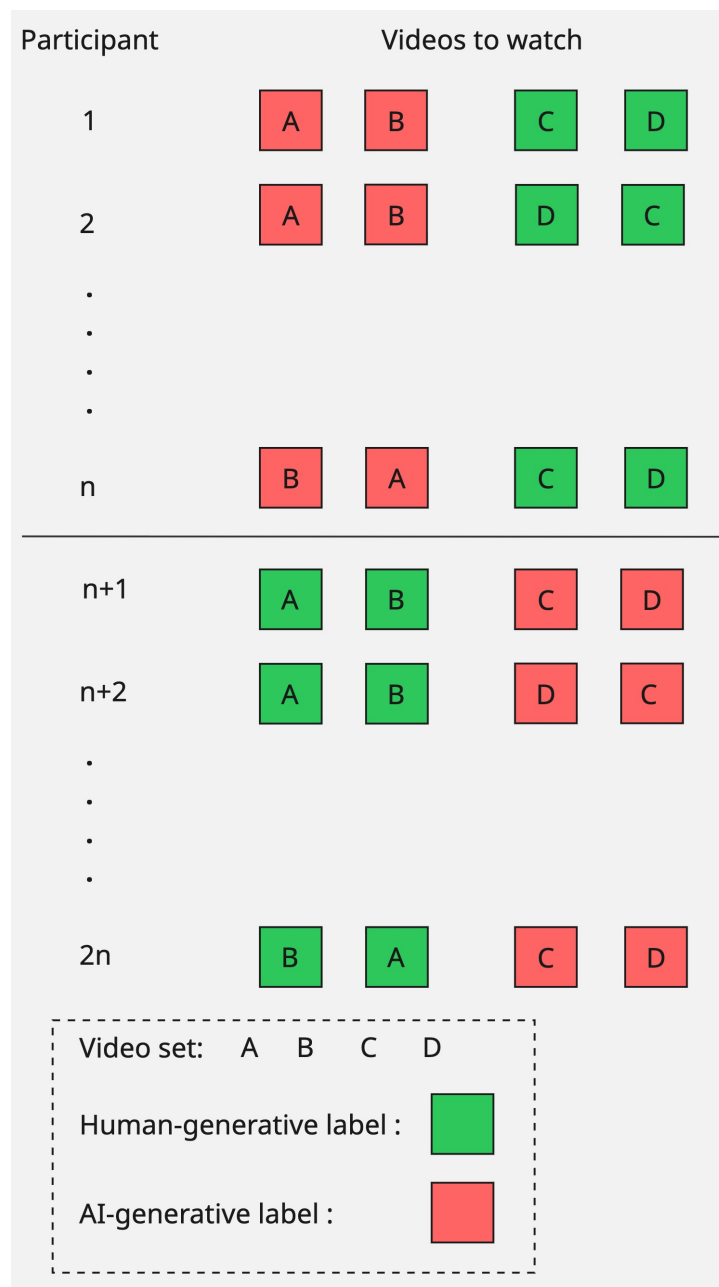


Figure 1: Visualization of the experiment design

A.4 Step-by-step procedure

- 1) **Introduction and consent:** Subject is informed on the experiment focus on their emotional arousal with AI and Human generated videos. They are asked to wear the HR sensor and to read and sign the forms in [A.10](#) [A.9](#).
- 2) **Sensor placement:** HR sensor attach time is noted in a file. From now, the subject only interacts with the PsychoPy interface on the experimenter computer.
- 3) **Trial setup loading:** The program would load the setup for the specific participant, selecting the order and the belief (AI or Human) of the videos to show.

- 4) **Instructions:** Subject is instructed about the videos they will see and they are asked to relax for two minutes to avoid fluctuations.
- 5) **Task:** Subject is presented with the cue for the video origin, then a fixation cross appears. Then the video would play for its entire duration, while the heart rate was being recorded.
- 6) **SAM and focus ratings:** After the video, the subject is asked to answer to the SAM to provide valence, arousal and dominance ratings. They are also asked to rate their focus level on the content of the video on a similar scale. The last two steps repeated for all the (four) videos they were expected to watch.
- 7) **Attitude towards AI rating:** Subject rated their prior opinion of AI-generated videos on a five-point scale.
- 8) **Debriefing:** HR sensor de-attach time is noted in a file. The subject's oral answers to whether they trusted the authenticity cues was noted as well. They were then informed that some cues had been intentionally forged as the purpose of the study was to examine how this belief influenced their emotional responses.

A.5 Script read to the subject

Hi, and welcome!

Thank you for joining our study. We're students from Experiments in Cognitive Science, our names are Lanlan, Francesco and Jasmin. For this experiment we're studying how people's heart rates respond while watching different videos. Before we start, you'll need to wear a small portable heart rate sensor on your chest. This sensor will record your heartbeat throughout the experiment. It's completely safe and comfortable — just make sure it stays in place and avoid touching it during the session. Before the experiment starts, we will record a 2-minute baseline of your heart rate. If you have either a pacemaker or a heart transplant, we recommend that you do not participate in this experiment.

You'll watch four short, content-friendly videos — nothing violent or uncomfortable. Between each video, there will be a short break, and after each video, you'll fill out a short questionnaire about your experience with this video. Each video will have a label which either says "AI-generated" or "human-generated".

You can stop and withdraw at any time if you feel uncomfortable or simply don't want to continue - you don't have to give us a reason why.

If you have a heart transplant or pacemaker you can unfortunately not do this experiment, since it goes against the guidelines of using the Firstbeat ECG-device.

If you have any questions, please feel free to ask before we start. Thank you again for taking part of the experiment!

A.5.1 Debriefing

Thanks again for participating. In the beginning of this experiment we told you that there would be both AI-generated and human-generated videos. This was not true, since we had to maintain the integrity of the experiment. In reality all videos were real, human-generated videos and none of it was AI-generated. The experiment was actually about how prior attitudes towards AI would affect the emotional engagement in the video. Therefore we had to manipulate your belief, into thinking that the videos were actually AI-generated, in order to elicit a genuine response.

Now we have a question: Were you convinced that we told you the truth about the nature of the videos? Or did you not believe that the labels were correct? (*Remember to note this answer down for their datapoint, in case it ends up being an outlier!*)

A.6 Experiment setup

Participant	Order 1	Order 2	Order 3	Order 4
1	A _H	B _H	C _A	D _A
2	A _H	B _H	D _A	C _A
3	A _H	C _A	B _H	D _A
4	A _H	C _A	D _A	B _H
5	A _H	D _A	B _H	C _A
6	A _H	D _A	C _A	B _H
7	B _H	A _H	C _A	D _A
8	B _H	A _H	D _A	C _A
9	B _H	C _A	A _H	D _A
10	B _H	C _A	D _A	A _H
11	B _H	D _A	A _H	C _A
12	B _H	D _A	C _A	A _H
13	C _H	A _A	B _A	D _H
14	C _H	A _A	D _H	B _A
15	C _H	B _A	A _A	D _H
16	C _H	B _A	D _H	A _A
17	C _H	D _H	A _A	B _A
18	C _H	D _H	B _A	A _A
19	D _H	A _A	B _A	C _H
20	D _H	A _A	C _H	B _A
21	D _H	B _A	A _A	C _H
22	D _H	B _A	C _H	A _A
23	D _H	C _H	A _A	B _A
24	D _H	C _H	B _A	A _A

Table 1: Video order and assigned authenticity belief for each participant. Videos A-D correspond to the four experimental stimuli. The subscript _H=Human, _A=AI corresponds to the hint they were presented with.

A.7 Variables distribution

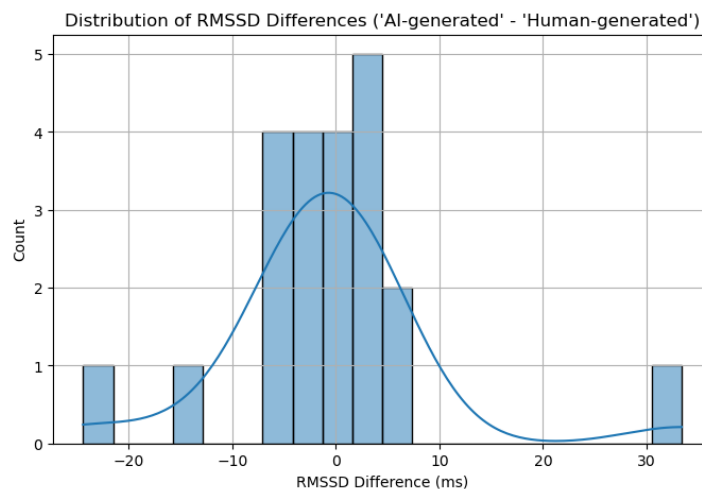


Figure 2: Distribution of data fitted to a normal distribution.

Test	W	p-value	r	"AI" 95% CI	"Human" 95% CI
HRV	102	0.4434	-0.170	[15.473, 60.439]	[18.141, 52.567]
Valence	34	0.0123	-0.640	[3.000, 3.500]	[3.500, 4.500]
Arousal	58	0.0735	-0.474	[2.500, 3.250]	[2.000, 2.500]
Dominance	62.5	0.7702	-0.443	[2.250, 3.000]	[2.500, 3.250]

Table 2: Summary of Wilcoxon Signed-Rank Test Results with 95% Confidence Intervals.

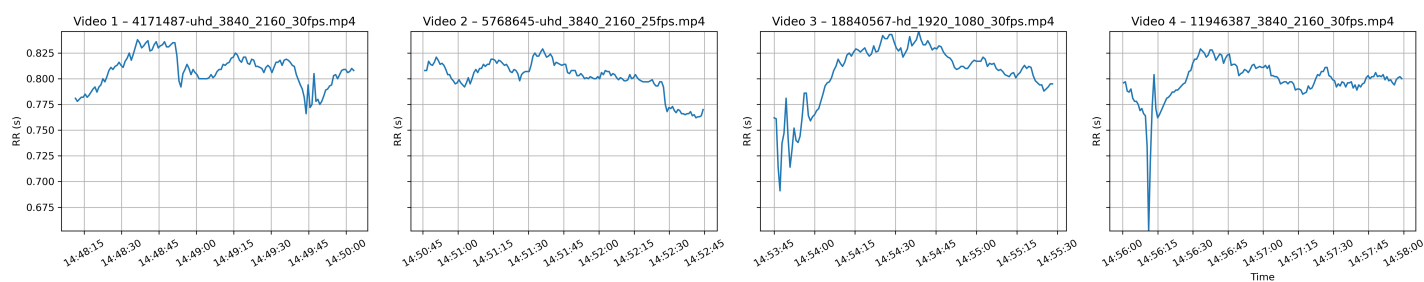


Figure 3: RR-intervals separated into each video for participant 1.

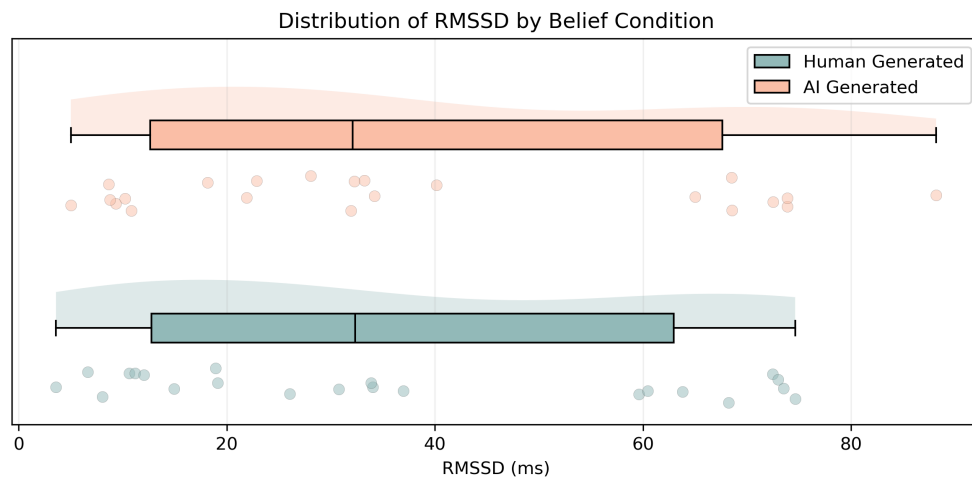


Figure 4: Distribution of the RMSSD data in a raincloud plot.

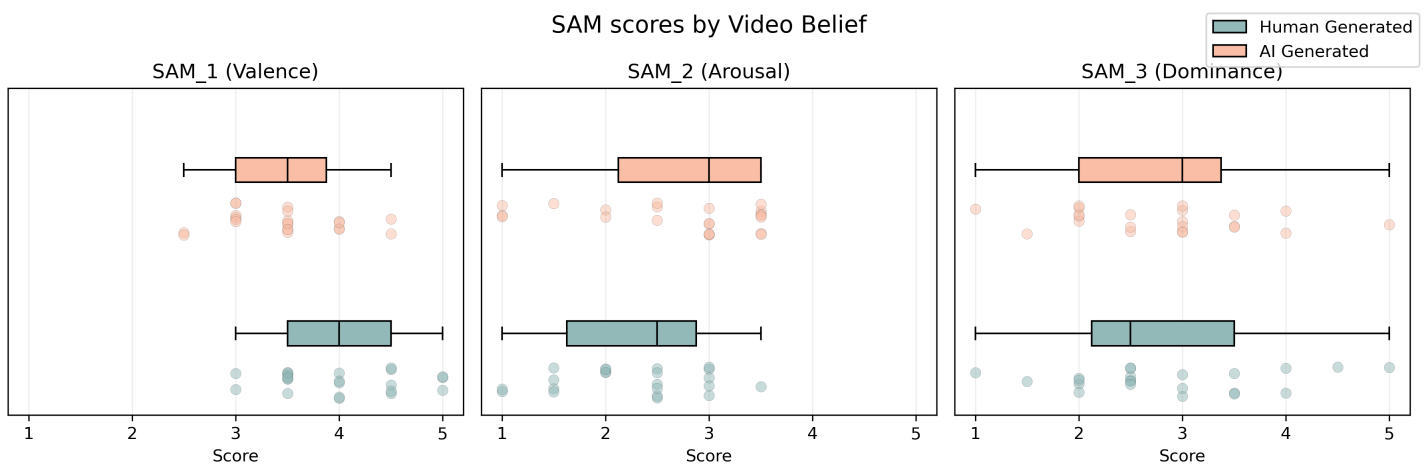


Figure 5: Distribution of the SAM response data in a boxplot.

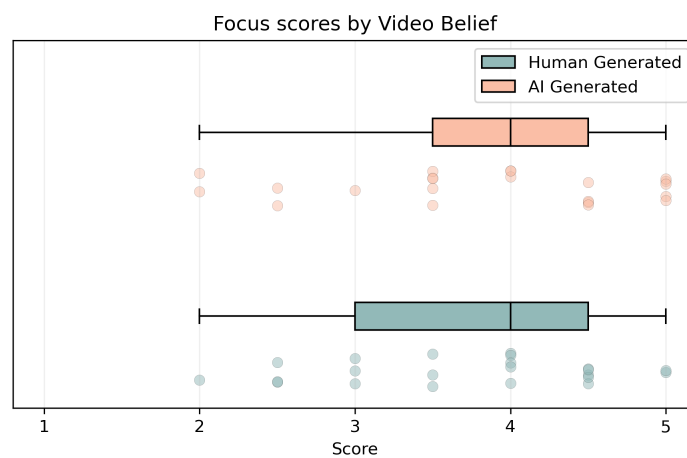


Figure 6: Distribution of the focus response data in a boxplot.

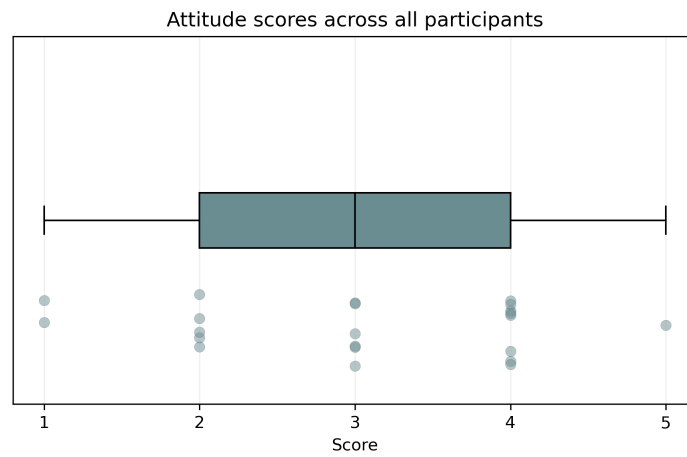


Figure 7: Distribution of the attitude towards AI response data in a boxplot.

A.8 Notes for each participant

Participant	Notes
1	Did not feel like videos were AI-generated while watching, felt more focused when watching presumed AI-videos in order to spot whether it was AI.
2	Was trying to figure out if it was AI or not, didn't know if the real videos were actually AI.
3	Speculated if videos were either all AI or all human, did not believe labels.
4	Believed labels, were only a bit sceptical, squirrel looked like AI.
5	Was a bit sceptical with some of the videos. Didn't believe otters were AI or squirrel was real.
6	Believed all the labels.
7	All videos felt real, although them being slowed down was an hint of AI.
8	Found it intriguing but expected the procedure as they did a similar experiment for a PhD. Believed all the labels except for one.
9	Fascinated as personally into heartrate and AI. More focused on videos with AI cue to check if it was true. Bored with the otters and happy with the dogs. He wouldn't have been able to distinguish the real and the AI if not for the label.
10	Focused on verifying if the ones labeled AI were real only at the very last video, where he still believed to the label (AI).
11	Didn't completely believe all the labels, though maybe ants were AI when it said human etc.
12	Believed what we said about the labels, but thought some of the labels didn't seem like they matched.
13	Didn't believe the labels, didn't believe that the videos that said AI-generated was actually AI-generated.
14	Video felt a bit too long and distracting. They were cute. Didn't focus that much on proving if they were lied to, but the ants video seemed not AI at all.
15	Thought that either all videos were AI or all videos were Human.
16	Was sceptical about video labels, some said human but thought they were AI
17	Did not really believe the labels, all videos looked like only AI videos.
18	Thought human videos were slow paced, and AI was fast paced, didn't believe labels for all videos
19	Didn't believe the labels that much, felt like labels were switched.
20	Felt like some labels were not human generated, could not differentiate nature of videos, believed it when said was AI generated.
21	Believed labels in some cases, but didn't with others.
22	Believed most of the labels.
23	Felt like the videos were relaxing, they liked it. About the labels, they felt inverted. Overall focused on the videos content.

Table 3: Notes from debriefing with each participant.

A.9 Information sheet

Project title: Emotional arousal and AI generation of videos

Participant information letter

Dear participant,

In this letter we will inform you about the purpose of the experiment and the procedures. It is important that you read the letter carefully. If you have any questions, do not hesitate to contact us (Francesco Balducci - s250200@student.dtu.dk) for clarification.

Your rights as participant

Your participation in this experiment is voluntary. This means that you can leave the experiment at any point in time without consequences for yourself, and without having to give a reason. We will ask you for an informed consent to participate after you have been informed about the experiment.

Purpose of the research project

In this research project we are investigating the possible effects on emotional response of various videos. Participants will be shown videos while we measure heart rate (HR) to evaluate physiological response and then assess the subjective feeling through questionnaires. The participant will be shown videos with labels "AI-generated" or "human-generated" throughout the experiment.

Data storage and handling

All recorded data - heart rate and responses to questionnaires - will be anonymized, which means that it will be stored without a connection to your name, address, CPR number, or any other information that would allow to directly identify you. For this, your data will be stored under a handle. Since heart rate data is considered biometric, it can only be pseudo-anonymized, meaning that it will not be stored with any personal information, but it may still be possible to identify you based on the raw data alone.

Heart rate physiological measurement

We will measure your heart rate (HR) as an indicator of your physiological state. Heart rate refers to the number of times your heart beats per minute. It is a non-invasive and safe method that provides information about your physical and emotional state. This data will be recorded using a Firstbeat Bodyguard 2 monitor. The device is attached to the skin with two electrodes and starts recording automatically when attached. This method does not influence your body in any physical way.

It is important that you remain as still and relaxed as possible during the experiment to avoid movement artifacts that can disturb the measurements. There will be regular breaks while answering questionnaires, during which you can move more freely. The study will take approximately 15-20 minutes in total.

Contact us

If you have any questions about the experiment, the methods or your safety and rights, do not hesitate to contact us by email (s250200@student.dtu.dk) or telephone (+45 71 53 95 90).

With kind regards,

Lanlan Yan, Francesco Balducci and Jasmin L. Aa. Petersen

A.10 Informed consent form

Experiment: _____

Investigator: _____

I confirm that:

- I was satisfactorily informed about the study concerned both verbally and in writing by means of the participant information letter.
- I have had the opportunity to put forward questions regarding the study and that these questions have been answered satisfactorily.
- I have carefully considered my participation in the experiment.
- I participate of my own free will.
- I am between 18 and 60 years of age, and am fluent in the language used in the experiment (English).
- I do not have any prior familiarity with the exact content of the experiment.
- I do not have a condition that can affect the outcome of the experiment, such as ADHD or heart-related diseases.

I agree that:

- My data will be acquired and stored for scientific purposes as mentioned in the participant information letter in pseudo-anonymized form.
- My pseudo-anonymized data will be stored and may be shared through a private repository.

I understand that:

- My participation is voluntary, and I have the right to withdraw from the experiment at any time without having to give a reason.
- My privacy is protected according to Danish law and European guidelines (GDPR; EU 2016/679).

I give my consent to take part in this experiment:

Full name

Date, Place

Signature

B References

- [1] S. Roy and S. Roy. “Deepfakes: Turning reality into chaos.” 2024, Springer. [Online]. Available: https://doi.org/10.1007/978-981-97-8457-8_12.
- [2] S. Guo, B. Swire-Thompson, and X. Hu. “Specific media literacy tips improve ai-generated visual misinformation discernment.” 2025, SpringerOpen. [Online]. Available: <https://doi.org/10.1186/s41235-025-00648-z>.
- [3] P. Núñez-Cacho, G. Mylonas, A. Kalogeras, and V. Molina-Moreno. “Exploring the transformative power of ai in art through a circular economy lens. a systematic literature review.” 2024, Elsevier. [Online]. Available: <https://doi.org/10.1016/j.heliyon.2024.e25388>.
- [4] B. Chesney and D. Citron. “Deep fakes: A looming challenge for privacy, democracy, and national security.” 2019, HeinOnline. [Online]. Available: <https://lawcat.berkeley.edu/record/1136469?v=pdf>.
- [5] M. Hameleers and F. Marquart. “It’s nothing but a deepfake! the effects of misinformation and deepfake labels delegitimizing an authentic political speech.” 2023. [Online]. Available: <https://ijoc.org/index.php/ijoc/article/view/20777>.
- [6] Y. Yang, T. Davis, and M. Hindman. “Visual misinformation on facebook.” 2023, Oxford University Press. [Online]. Available: <https://doi.org/10.1093/joc/jqac051>.
- [7] D. Gamage, K. Sasahara, and J. Chen. “The emergence of deepfakes and its societal implications: A systematic review.” 2021, TTO. [Online]. Available: <https://api.semanticscholar.org/CorpusID:244716544>.
- [8] A. Eiserbeck, M. Maier, J. Baum, and R. A. Rahman. “Deepfake smiles matter less - the psychological and neural impact of presumed ai-generated faces.” 2023, Nature. [Online]. Available: <https://www.nature.com/articles/s41598-023-42802-x>.
- [9] L. Arango, S. P. Singaraju, and O. Niininen. “Consumer responses to ai-generated charitable giving ads.” 2023, Taylor & Francis. [Online]. Available: <https://doi.org/10.1080/00913367.2023.2183285>.
- [10] U. Agudo, M. Arrese, K. G. Liberal, and H. Matute. “Assessing emotion and sensitivity of ai artwork.” 2022, Frontiers Media SA. [Online]. Available: <https://doi.org/10.3389/fpsyg.2022.879088>.
- [11] J. D. Brüns and M. Meißner. “Do you create your content yourself? using generative artificial intelligence for social media content creation diminishes perceived brand authenticity.” 2024, Elsevier. [Online]. Available: <https://doi.org/10.1016/j.jretconser.2024.103790>.
- [12] M. Bilucaglia et al. “Emotional reactions to ai-generated images: A pilot study using neurophysiological measures.” 2024, Springer. [Online]. Available: https://doi.org/10.1007/978-3-031-82487-6_11.
- [13] A. Minotakis. “Concerned, unprepared and burdened by the past: Generative ai in greek newsrooms.” 2025, Taylor & Francis. [Online]. Available: <https://doi.org/10.1080/17512786.2025.2553188>.
- [14] J. P. Cardenuto et al. “The age of synthetic realities: Challenges and opportunities.” 2023, APSIPA Transactions on Signal and Information Processing. [Online]. Available: <https://doi.org/10.48550/arXiv.2306.11503>.

- [15] F. Shaffer and J. P. Ginsberg. “An overview of heart rate variability metrics and norms.” 2017, PubMed Central. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5624990/>.
- [16] Kubius. “Heart rate variability normal range.” 2025, Kubius. [Online]. Available: <https://www.kubios.com/blog/heart-rate-variability-normal-range/>.
- [17] S. Kingdom. “Shapiro wilk test.” 2025, Statistics Kingdom. [Online]. Available: https://www.statskingdom.com/doc_shapiro_wilk.html.
- [18] S. Kingdom. “Wilcoxon signed-rank test.” 2025, Statistics Kingdom. [Online]. Available: https://www.statskingdom.com/test_wilcoxon_signed_ranks.html.